System Software - Operating Systems

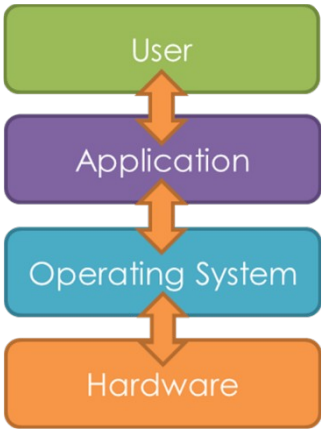
What is system software?

System software is software which allows a computer to run and to be maintained. "Operating System" software and "Utility Software" are the two main types of system software.

The Operating System

The operating system is the most important piece of software on any computer. Without it, no programs will run. This is because an operating system is responsible for controlling / communicating with the computer hardware. It provides a platform on which games, browsers, music players etc, can all work.

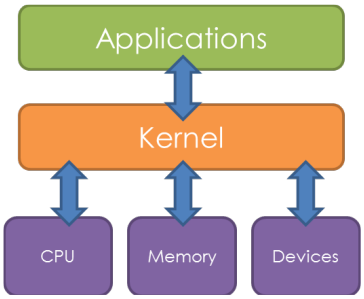
If you were to run an everyday program (e.g. a word processor), without an operating system, nothing would be displayed on the screen, nothing could be sent to the printer, nothing could be typed. This is because application software, does not know how to TALK to hardware devices. However, the operating system does. The operating system can also talk to the application that is running. So when you print a document, the application talks to the OS, which in turn talks to the printer.



The operating system sits between the user's applications and the hardware. It enables applications to use the hardware resources.

The Kernel

The kernel is the heart of the operating system and is responsible for looking after "the most low-level hardware operations". It is the kernel that applications make use of when they want to operate the computer's hardware.



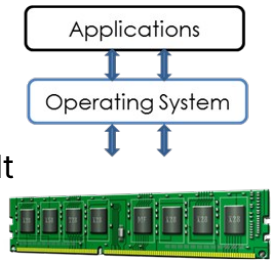
The 5 Major Roles of the Operating System

Ultimately, the operating system provides 5 major roles:

- Memory Management
- Peripheral Management
- Multi-Tasking
- Security
- User Interface

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Memory Management



The OS manages the RAM. When you load an application, the OS will copy the application to the RAM. In addition, the OS will ensure that each program that loads has its own space on the RAM. This stops other programs reading and writing to the memory area for another application. It is also responsible for managing virtual memory.

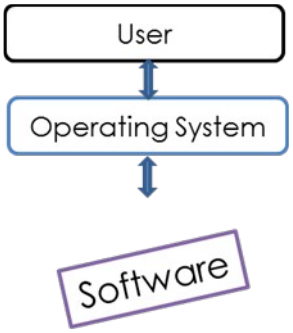
Peripheral Management



or

Peripherals are devices that are attached to a computer such as a printer speakers. The operating system controls these devices and gives programs access to them. When you print a document, the program doesn't know how to talk to the printer, it doesn't even know if one is installed. Instead it passes this data to the operating system which in turn passes it to the printer. The same happens when you save a file. The program passes this information to the operating system and the operating system is what saves files onto the hard disk.

The User Interface



The word interface means 'coming together'. In computers, an interface enables a user to interact with a computer. Operating Systems provide users with an interface so that they can interact with the computer. Operating systems will usually offer one of three major types of interface:

- Graphical User Interface (GUI)
- Menu Driven Interface
- Command Line Interface

Graphical User Interfaces



Uses windows, icons, menus and pointers (WIMP) to control the computer. Only been around since the mid 80s. Windows and Apple Mac OSx are examples of this kind of interface.

Menu Driven Interface



Uses menus to control the computer. These were very popular on early mobile phones and are seen in all high streets on cash machines and also on many MP3 players. Only offer limited functions (ie. if it is not on the menu then you can't do it!)

Command Line Interface

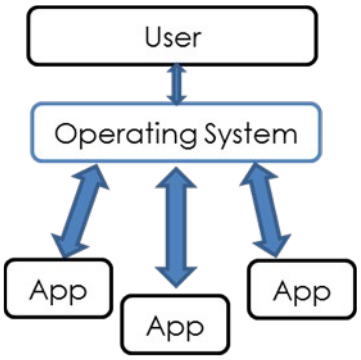


Used to be the only way to interact with a computer (pre-80s) and is still used today by 'Linux users' and other professionals with technical knowledge. With this kind of interface 'written commands' control computers.

Multi-Tasking

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As we know, a CPU's core works one process at a time. An operating system enables multi-tasking by managing the CPU's processes so that several programs can run at once. On old computers, a CPU would focus on the processes of one program. On a multi-tasking operating system, many programs can run at the same time. The operating system enables this to happen by either allowing processes to take it in turn, or getting the shortest job done first. Because CPU's are so quick now-a-days, it appears to be simultaneous and allows many programs to be used at once.



Security

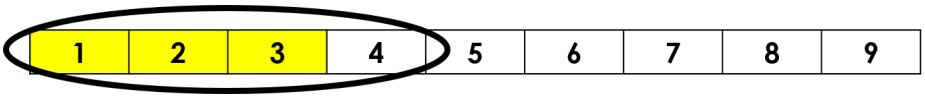
Often computers will be used by several users. Operating systems manage these users and ensure that a user's data is only seen by them and not by others. In this way an OS provides security to users. They give users 'usernames' and 'passwords' so that data is kept secure. They also give different users different permissions. This means that one user may be given permission to install new programs whereas others cannot. In this way, an operating system can ensure that a computer system is not open to abuse from any user.



Questions *(The question zone you choose must either match your target grade or be higher!)*

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Question Zone 1-3



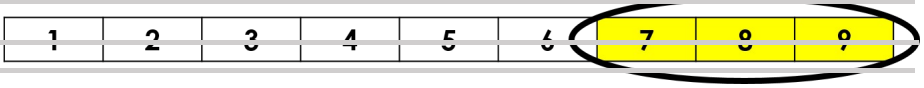
1. Define the term 'System Software'. [2]
2. "The Operating System sits between the user's applications and the hardware". What is meant by this statement? [3]
3. Briefly describe the 5 major roles of an 'Operating System'.[5]

Question Zone 4-6



1. Briefly describe the 5 major roles of an 'Operating System'.[5]
2. Explain how the OS manages memory. [4]
3. Explain how the OS enables multi-tasking. [4]

Question Zone 7-9



1. Explain how the OS enables multi-tasking. [4]
2. Explain the role of the 'Kernel' in an 'Operating System'. [2]
3. Discuss the need for the 3 major types of interface. [6]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Operating System (OS): *Software, which manages the hardware and software resources of a computer system and provides an interface for the user.*

Kernel: The heart of the operating system, responsible for looking after "the most low-level hardware operations".

Multi-Tasking*: The ability of an OS to supply the CPU with instructions from multiple programs in particular orders to create the illusion that a computer is processing multiple programs simultaneously.*

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Stick answer sheet here

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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